

Autonomous Salvage ROVs in the Hadal Research Base “Abyssus”

FREE CHOICE ASSIGNMENT FOR THE COURSE *AUTOMATED PLANNING*, UNIVERSITY OF TRENTO, A.Y. 25/26

Stefano Racca,
stefano.racca@unitn.it

Giorgia Lorengo
giorgia.lorenco@unitn.it

1 Introduction

This document aims to describe the **Free Choice Scenario** for the course *Automated Planning: Theory and Practice* at the University of Trento. Specifically, the project focuses on planning the actions of **autonomous robots tasked with managing an underwater salvage operation in a damaged research base**.

2 Scenario

Located in the Mariana Trench, the **Abyssus Base** stores extremophile biological samples. The peculiarity of this base lies in the fact that it is almost entirely submerged internally and operated exclusively by robotic systems. Due to the extreme depth, most of the facility is intentionally kept flooded, allowing the structure to better withstand external pressure and reduce long-term mechanical stress.

All operations within the base are carried out by ROVs, which move through the water-filled modules to perform research, transport materials, and handle recovery tasks. Only a limited section of the station is maintained as an on-filled environment, primarily to host pressure-sensitive equipment and enable controlled interactions when required.

Unfortunately, a hydrothermal eruption has caused severe structural damage in one of its area. Autonomous **Remotely Operated Vehicles (ROVs)** must secure and relocate the samples before a total structural collapse occurs.

The base is organized into the following **specialized modules**:

- **Docking Station:** The initial location of all ROVs, where missions are deployed and vehicles can recharge.
- **Research Wings (Alpha & Beta):** The Abyssus Base is divided into two research wings. **Wing Beta** remains stable, while **Wing Alpha** has become isolated due to structural damage and is now accessible only through a narrow breach.
- **Transfer Zone:** A secure hub connecting the

internal corridors of Wing Beta to the structural breach. It serves as a coordination and handover point between different classes of ROVs.

- **Decompression Chamber:** A sealed interface module that enables safe transfer between flooded (water) zones and dry (air-filled) sections of the base. It operates by draining or flooding the chamber to equalize environmental conditions, allowing ROVs and materials to pass without compromising structural integrity or internal atmosphere. This module is essential for maintaining separation between aquatic and habitable areas.
- **Pressure Stabilizer:** A controlled environment required to stabilize pressure-sensitive samples before transferring them from Wing Alpha to the Bio-Vault. This is optional, depending on the type of object that has been rescued.
- **Bio-Vault:** The final secure storage destination for all salvaged samples.

In the ocean, danger is ever-present. ROVs must therefore account for **Marine Hazards** during the motion planning phase. These include:

- **Small Fish Schools:** These populate the flooded tunnel that connects the Transfer Zone to the Decompression Chamber, increasing traversal time and energy consumption.
- **Apex Predators:** Large marine creatures patrol the area outside the breach, sometimes entering the leftovers of the Alpha Wing, creating temporary *no-go* zones for smaller ROVs.

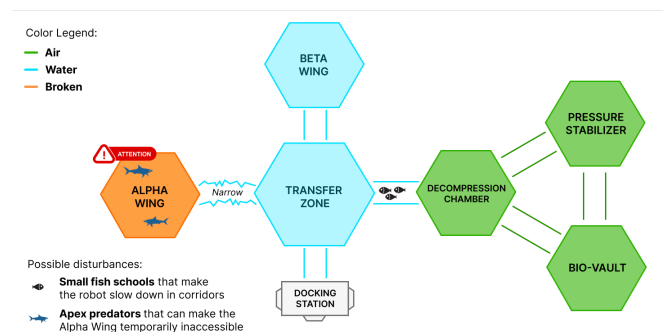


Figure 1: Schematic representation of the environment.

To accomplish the mission, two **types of agents** are deployed:

- **Small-Class ROVs:** Agile units capable of passing through the structural breach to access Wing Alpha. They have a limited carrying capacity (1 unit).
- **Large-Class ROVs:** High-capacity transport units (up to 4 units) restricted to the main internal corridors and the Transfer Zone.

3 Assignments and Subtasks

To address the overall problem, the assignment is divided into five sequential subproblems, defined as follows.

3.1 Task 1: Single ROV Problem

The objective is to retrieve samples and transport them from Wing Alpha to the Bio-Vault, passing through the Decompression Chamber if required by the type of object that is being secured.

Initial state: Samples are distributed across the research wings according to a reproducible random assignment. A single ROV is initially located at the Docking Station.

Goal: Transport all samples to the Bio-Vault, possibly via the Decompression Chamber when required.

3.2 Task 2: Multi-Agent Coordination

A single robot is insufficient to retrieve all samples from Wing Alpha. This task introduces multiple small and large ROVs, with the characteristics described in section 2.

The main challenge lies in handling **capacity constraints** and coordinating the transfer of samples at the Transfer Zone.

3.3 Task 3: Hierarchical Task Networks

The same domain is now addressed using **Hierarchical Task Networks (HTN)**. This task requires decomposing high-level objectives into structured subtasks for both ROV classes.

3.4 Task 4: Temporal Planning

At this stage, the problem increases in complexity. Actions are no longer instantaneous: their duration may vary depending on environmental conditions (e.g., traversal through fish schools requires more time).

The challenge involves reasoning about **durative actions**, temporal constraints, and parallel execution.

3.5 Task 5: PlanSys2 and ROS2

The final problem consists in implementing within the PlanSys2 the last problem resulting as outcome of subsection 3.4 using fake actions.